Cross-dataset 3 Results and Analysis

We train and test our dataset for:
GRU (Gated Recurrent Unit), Transformer and GAN (Generative Adversarial Network)

Train Dataset: schedule_predictions.csv
Test Dataset: feb2026rb2r2_updated.txt

Summary:
For this dataset, all models perform poorly, with negative R^2 for both voltage and power, indicating failure to learn meaningful relationships. For voltage, GRU and Transformer perform similarly and slightly better than GAN, but still lack predictive reliability. For power, all models show extremely high errors and very negative R^2 (~ -3), suggesting severe difficulty in capturing system behavior. **This indicates the dataset is highly challenging (likely noisy or distribution-shifted), where neither temporal nor attention-based models generalize well, and GAN offers no advantage.**

Voltage GRU Results:
MAE: 0.28466735270481514
RMSE: 0.35664668319828113
R2: -0.12299029571744158

Voltage Transformer Results:
MAE: 0.2820503705049771
RMSE: 0.35299867381405664
R2: -0.10012835592593383

Voltage GAN Results:
MAE: 0.27885881401508666
RMSE: 0.3639430064129095
R2: -0.16940236519491658

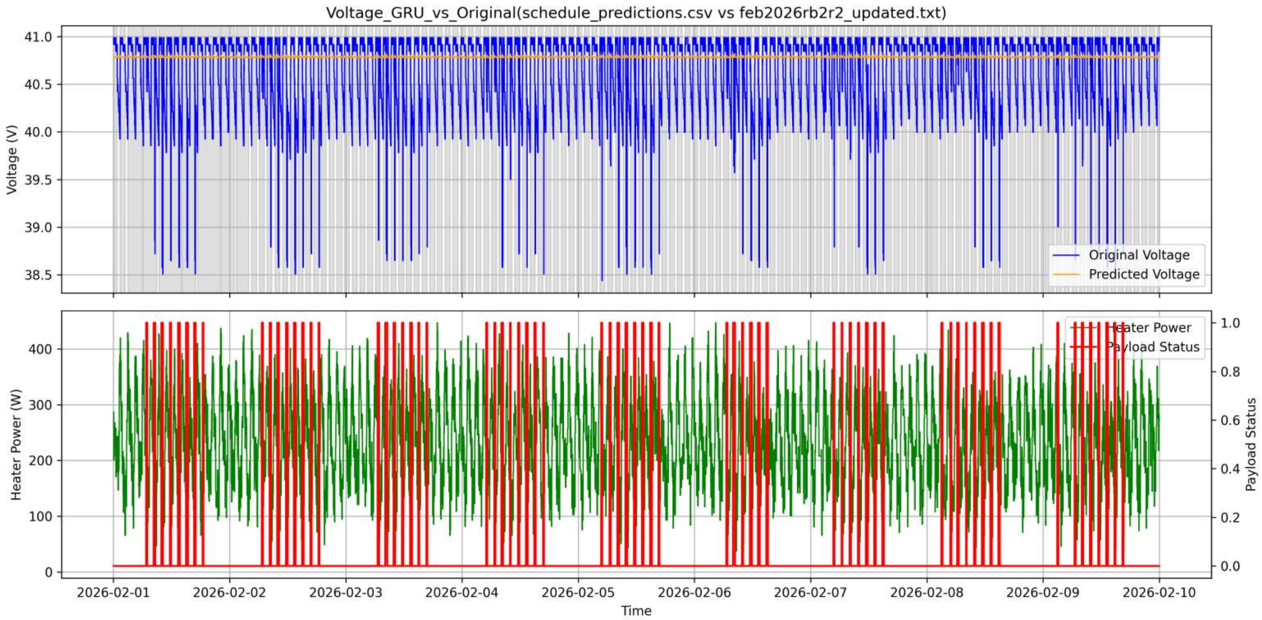
Power GRU Results:
MAE: 266.33897886776884
RMSE: 292.28110352307795
R2: -3.302467900362166

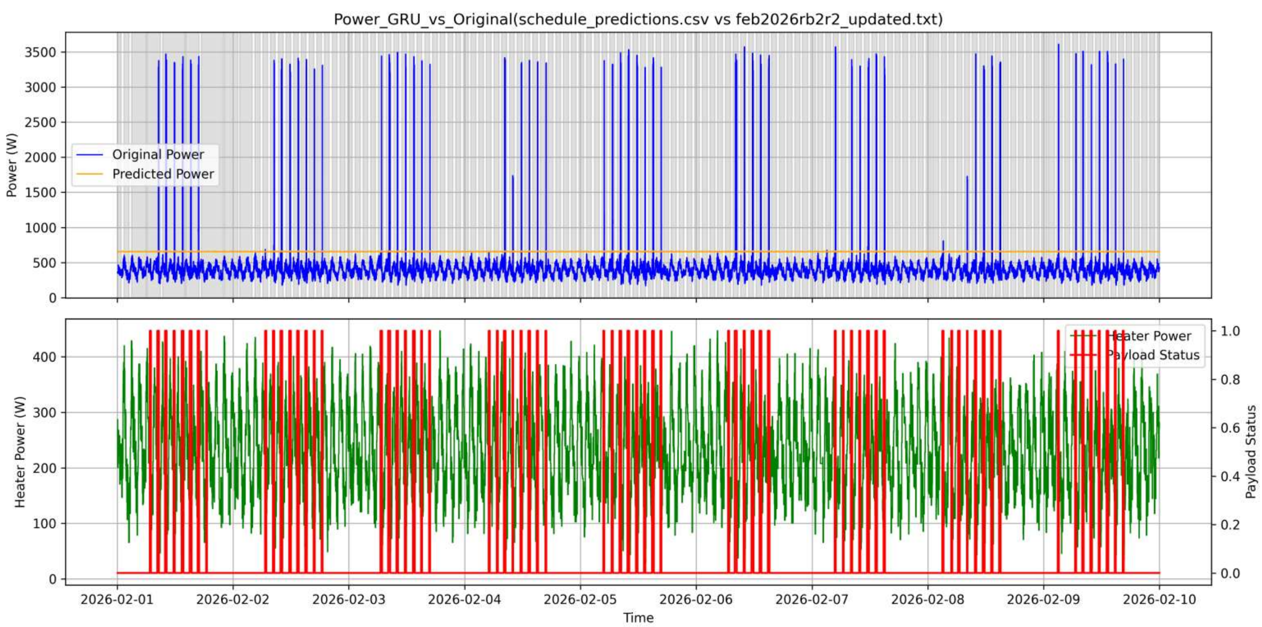
Power Transformer Results:
MAE: 266.5638673026016
RMSE: 292.4821674913774
R2: -3.3083133584750932

Power GAN Results:
MAE: 262.4253361114288
RMSE: 288.86816893128525
R2: -3.202501485509506

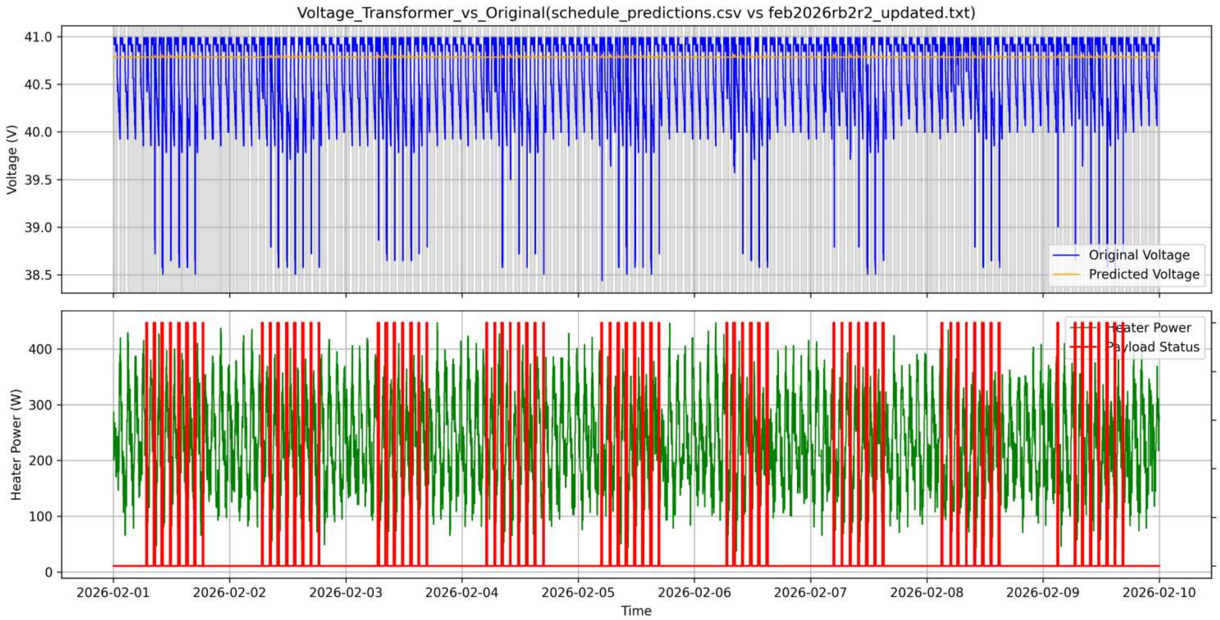
Plots:

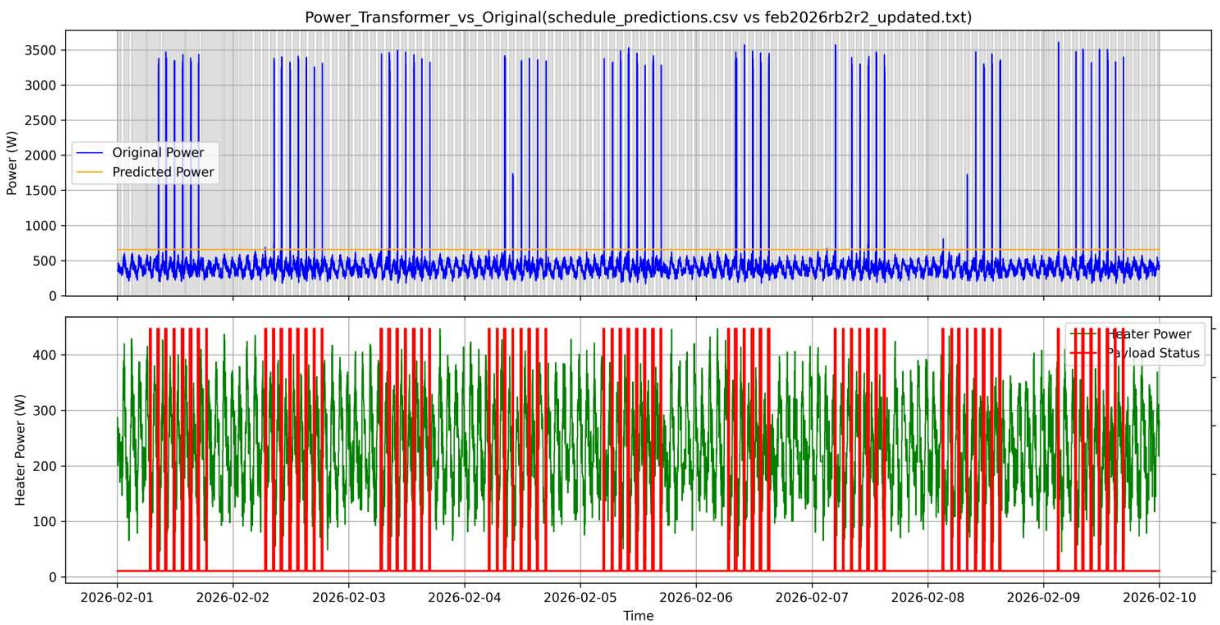
1. GRU





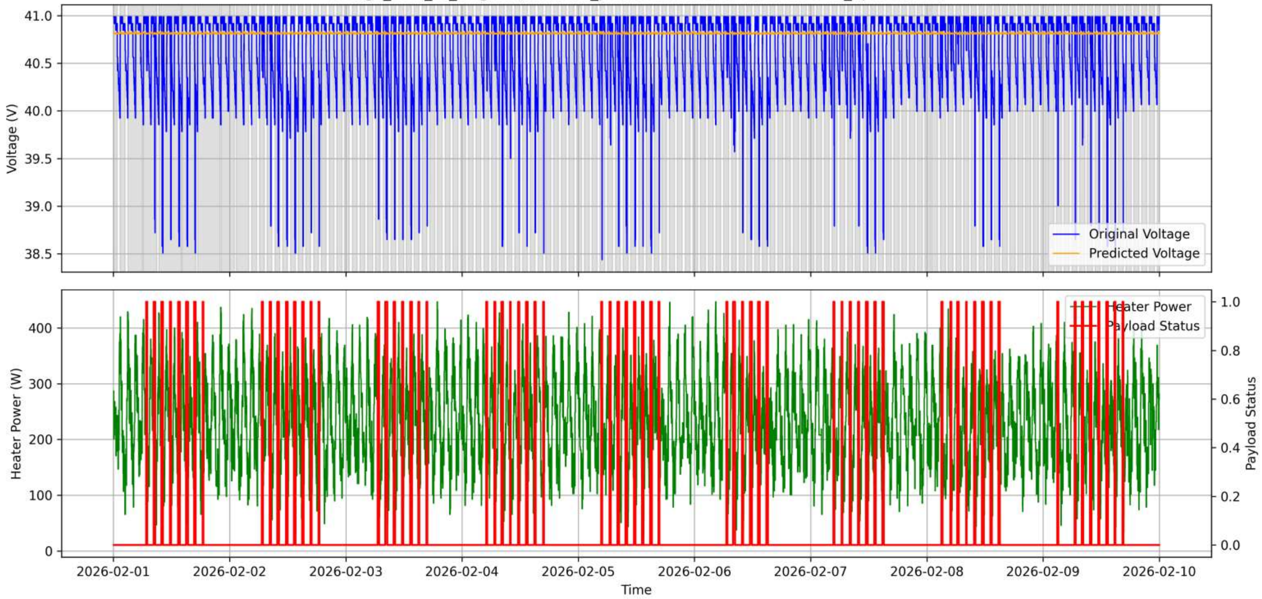
2. Transformer


Voltage_Transformer_vs_Original(schedule_predictions.csv vs feb2026rb2r2_updated.txt)
Voltage (V)
Original Voltage
Predicted Voltage
Heater Power (W)
Payload Status
Heater Power
Payload Status
Time


Power_Transformer_vs_Original(schedule_predictions.csv vs feb2026rb2r2_updated.txt)
Power (W)
Original Power
Predicted Power
Heater Power (W)
Payload Status
Heater Power
Payload Status
Time

3. GAN

Voltage_GAN_vs_Original(schedule_predictions.csv vs feb2026rb2r2_updated.txt)



Power_GAN_vs_Original(schedule_predictions.csv vs feb2026rb2r2_updated.txt)